Early Detection of Interest Loss in Hebrew WhatsApp Sales Conversations: Textual, Behavioral, and Prefix-Aware Modeling

Early Detection of Interest Loss in Hebrew WhatsApp Conversations

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Reichman University | Natural Language Processing | 2026

Can we detect loss of interest while a conversation is still developing?

Secondary question: Do behavioral communication patterns provide information beyond text?

Task and Data

100% SYNTHETIC / LLM-GENERATED

3,055
synthetic Hebrew WhatsApp-style sales conversations

2 classes
Interested vs. Losing Interest (~balanced)

2,137 / 459 / 459
Train / Validation / Test split

Two complementary signal sources

TEXT

"What is being said?"

- Semantic content
- Intent
- Language cues

BEHAVIOR

"How is the interaction evolving?"

- Participation & message length
- Response timing
- Session / gap structure
- Temporal change over the conversation

From Classification to Early Detection

A — Full Conversation

Pure Behavioral → AlephBERT → AlephBERT + Behavior → Continued AlephBERT (control)

AlephBERT: Seker et al., 2022

B — Early Detection (Partial Conversations)

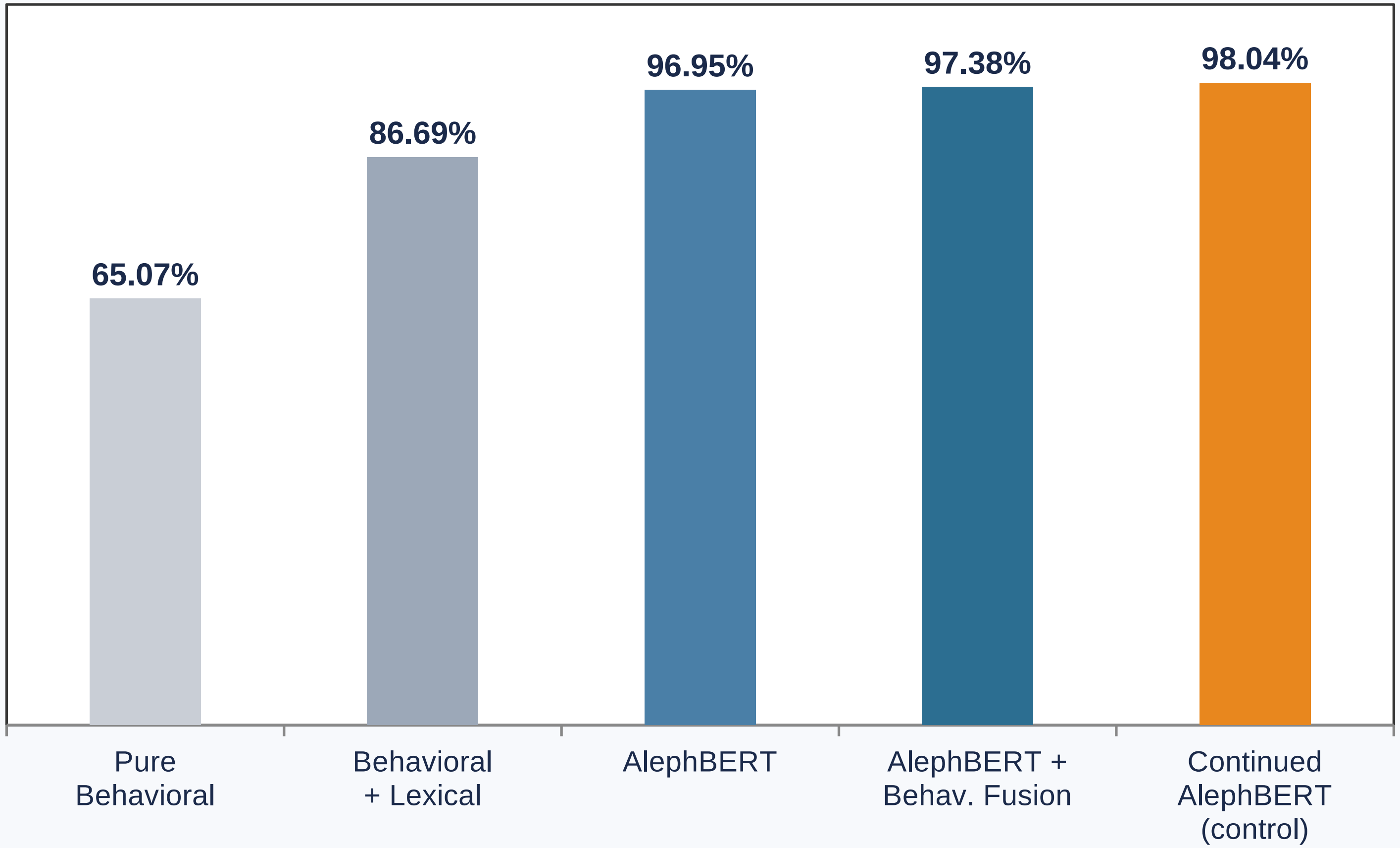
25%	50%	75%	100%

Conversation observed →

- **E1** Train on 100% of the conversation, evaluate on 25 / 50 / 75 / 100%
- **E2** Train directly on prefixes, text only
- **E3** Train directly on prefixes, text + behavioral features

Behavioral features use ONLY information visible at the current prefix — no future-message leakage.

Text Is Already Very Strong



97.38% vs **98.04%**

Fusion vs. Controlled Text-Only

Once training is controlled, text-only beats fusion.

Behavioral fusion does not provide a clear full-context advantage.

The continued-training control repeats Fusion's extra training steps, without adding behavioral features — and reaches 98.04%. Since an equivalently-trained text-only model outperforms Fusion, the apparent full-conversation gain cannot be attributed to behavior. This is exactly what a controlled ablation is for.

Early Detection Changes the Picture

AT 75% OBSERVED

51.14%

↓

86.93%

↓

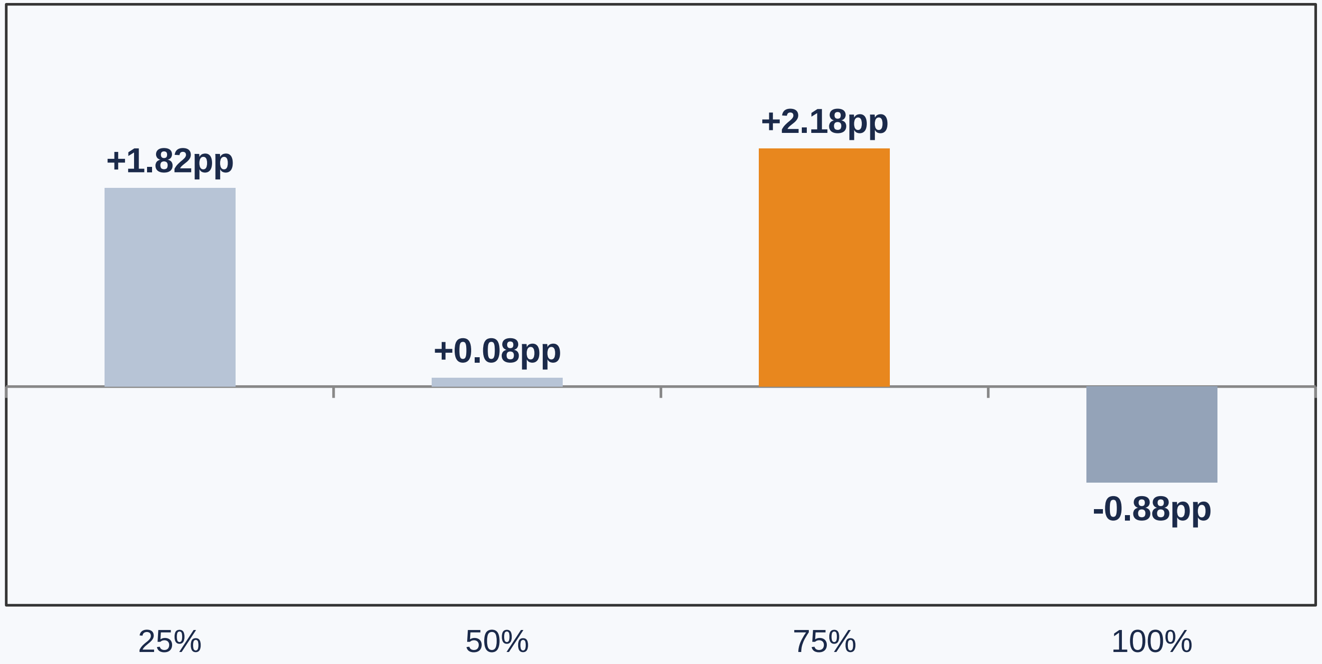
89.11%

Full-trained → Prefix-aware → Prefix-aware + Behavior

Prefix-aware training is critical for early detection.

E1 reaches 98.04% with the full conversation but collapses to ~33% early — strong full-conversation performance does not guarantee early detection. E2's prefix-aware training lifts 75% performance from 51.14% to 86.93% (+35.8 pp). E3 adds behavioral features, reaching 89.11%.

Does Behavior Add Information?



STATISTICAL EVIDENCE (75%)

Bootstrap 95% CI:
[+0.20, +4.36] pp
McNemar p = 0.0639

Strongest directional evidence at 75%, but not statistically conclusive under McNemar.

ERROR ANALYSIS AT 75%

17 fixed / **7 regressed**

net +10 correct out of 459

CLASS-ASYMMETRIC EFFECT

True Interested: net +11 | True Losing Interest: net −1

Behavior mainly reduced false losing-interest predictions. The benefit is not uniform across classes.

What We Learned

1. Behavioral dynamics alone contain useful predictive signal.
2. Full-conversation performance can substantially overestimate early-detection ability.
3. Prefix-aware training dramatically improves prediction under partial observation.
4. Behavioral fusion shows modest, stage-dependent complementary value — strongest at 75% — but does not establish a universal or statistically conclusive improvement.

OUR CONTRIBUTION

- Early-detection framing
- Controlled prefix evaluation
- Prefix-aware training
- Behavioral / text fusion
- Controlled ablation
- Paired statistical validation

Evaluation is based on synthetic Hebrew conversations; future work should validate the findings on real-world data and across multiple training seeds.